

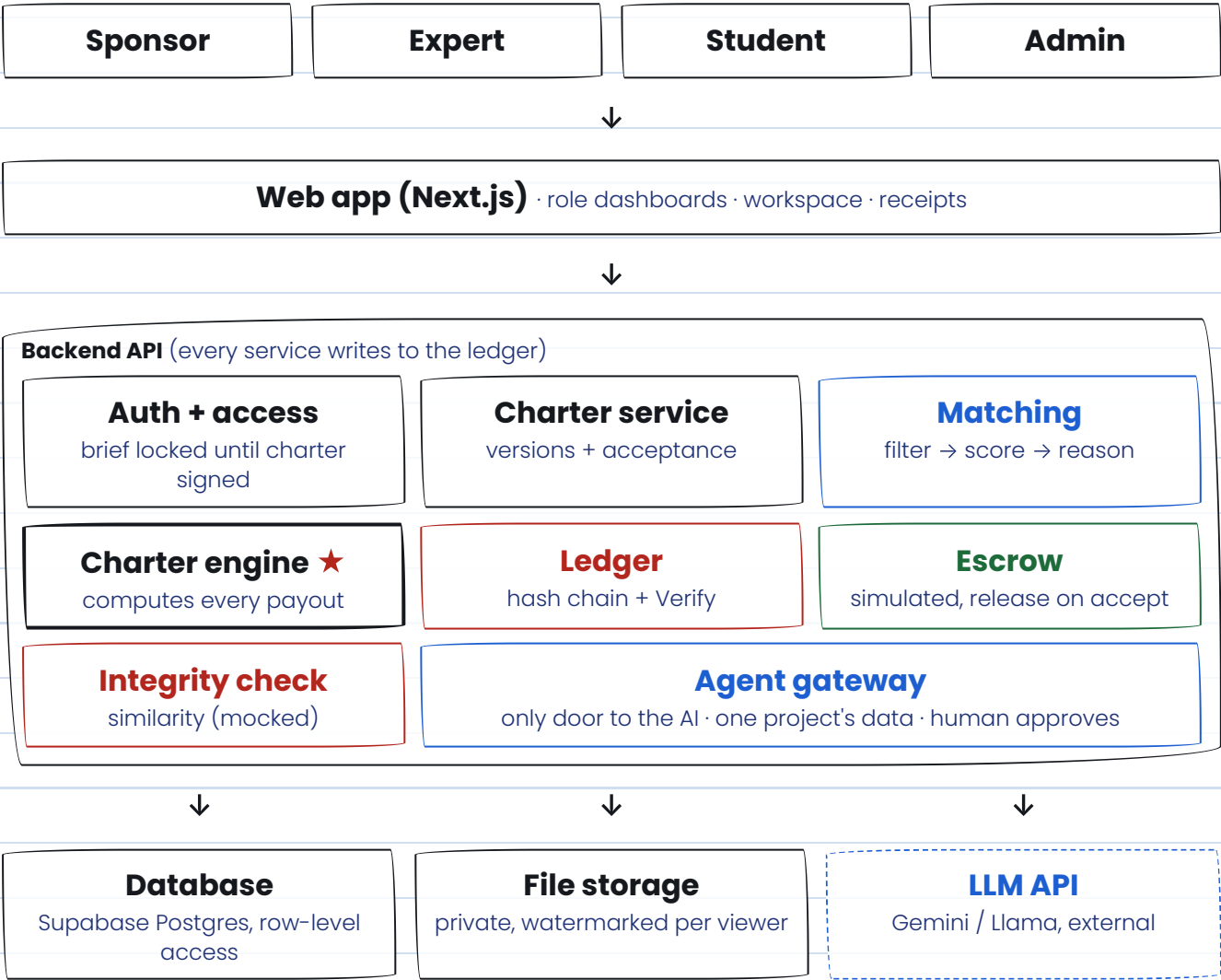
1 Our solution in short

Problem: sponsors, experts and students can't prove who did the work, terms change quietly, and AI blurs credit.

Idea: the project charter is **code, not a PDF**. One engine reads the charter + the ledger and pays everyone, with a **receipt for every rupee**.

Focus area: C · Fair rewards & governance (split engine, contribution scoring, disputes).

2 System architecture



Key: blue = AI · red = security · green = money · ★ = our innovation

3 Data model

User · role, verified, skills, hours	Project · sponsor, summary, brief (locked)
Charter · version, model, split rules	Milestone · criteria, amount, status
Contribution · author, file hash, AI share	Review · reviewer, verdict, impact 0–10
Ledger Entry · actor, on_behalf_of, event, prev hash, hash	Escrow · amount, funded / released / frozen
Payout · amount, receipt lines	Dispute · reason, evidence (ledger ids), outcome

Links: Project → 1 Charter per version → many Milestones → many Contributions → Reviews. Milestone → 1 Escrow → many Payouts. **Everything writes a Ledger Entry. on_behalf_of** = the human who owns every AI action.

4 Demo walk-through

Sponsor Dr. Anjali posts "Low-cost diabetic retinopathy detection on edge devices" for ₹1,00,000.

Step	What our system does
1 Post	Public summary + charter visible; confidential brief locked .
2 Scope	Scoping agent drafts 2 milestones + skills; sponsor approves.
3 Match	Expert + 3 students ranked with reasons ; Rohan filtered (works for a rival).
4 Join	Each member signs charter v1 → brief unlocks. Non-member still gets "denied".
5 Fund	₹1,00,000 locked in escrow before work starts.
6 Work	Humans + agents work; every action logged under a human owner.
7 Catch	Copied file scores 82% similar → flagged for expert review.
8 Accept	Sponsor accepts → engine splits → escrow pays , a receipt per person. Unpaid project → certificates + co-author credit.
9 Verify	Edit an old ledger entry → Verify fails at that entry.
10 Twist	Panel's corner case is added as an event → engine re-runs live.

5 Agent design

Agent	Owner	Can access	Human must approve
Scoping	Sponsor	Summary + brief	Publishing milestones
Matching	Platform	Profiles, ledger history	Sending invites
Coding / Research	Student	This project's files only	Adding work to project
Reviewer	Expert	Submission + criteria	Accept / reject



- **Prompt-injection defence:** documents are treated as data, never as instructions.
- **Sensitive projects** use only approved or self-hosted models.
- **AI doesn't earn:** compute cost comes from the 5% AI reserve; credit goes to the owner, with the AI share shown.

6 Ledger design

Entry contents: seq no · time · actor · on_behalf_of · event type · details · previous hash · own hash.

Hash chaining: hash = SHA-256(previous hash + this entry's contents).



Tamper detection: the **Verify** button recomputes every hash from #1; the first mismatch is shown in red ("broken at #2").

- **Append-only:** a database rule blocks edits; mistakes are fixed by a new *CORRECTION* entry.
- **Audit view** shows the full history to admins and members.
- **Later:** anchor the latest hash on a public blockchain.

7 Charter design & reward logic

Every charter states, before anyone applies: engagement model (funded · stipend · knowledge-sharing · institutional credit) · rewards · scope + milestones · roles · IP + publication · confidentiality + allowed AI tools · credit split · exit + disputes · future commercialisation.

- **Signed per version** + model acknowledged separately, recorded in the ledger.
- **Never changed silently:** a change is a new version; decline = leave with full credit.
- **Rules as code:** the engine runs these fields, so terms and payouts can't disagree.

Split formula (configurable):

distributable = budget - fee (10%) - AI reserve (5%)
expert = 30% × distributable · student pool = distributable - expert
student = (40% of pool ÷ no. of students) + (60% of pool × **weight**)
weight = share of expert-reviewed impact (not commit counts)

Worked example · funded, ₹1,00,000

Who	Weight	Equal part	Weighted part	Total ₹
Dr. Kiran (expert)	—	—	—	25,500
Priya	0.5	7,933	17,850	25,783
Arjun	0.3	7,933	10,710	18,643
Meera	0.2	7,933	7,140	15,074
Total (after ₹10,000 fee + ₹5,000 AI reserve)				85,000 ✓

Arjun's receipt: "Charter v1 · pool ₹59,500 · equal ₹7,933 + 60% × 0.3 = ₹10,710 · weight from reviewed work · AI share 70% disclosed."

Worked example · non-monetary (knowledge-sharing)

Same engine, budget ₹0 → each member gets a **certificate, co-author credit** and an expert recommendation, issued from their ledger record. Nothing on the page mentions pay.

8 Security mechanism per threat

Threat	Our mechanism
Plagiarism / undisclosed AI	Similarity check (mocked) + human review · AI-use declaration · agent actions auto-logged
Theft by insiders	Brief only after signing · watermark per viewer · no downloads · access logs
Theft by sponsor	Escrow before work · full files only on acceptance · ledger timestamps
Credit theft	File hash at submission + tamper-evident ledger
Fake accounts / wrong people	One verified ID · lead approval · conflict-of-interest declaration
AI data leak / prompt injection	Sensitivity labels · approved models · content = data · human approval
Bait-and-switch terms	Versioned charter · re-acceptance · ledger records version signed
Ledger / payment tampering	Append-only hash chain · corrections as new entries · audit view

9 Corner cases

- **Student quits midway:** silent 5 days → nudge → day 7 inactive → exit recorded. Credit kept, pro-rata share, access revoked; confidentiality clause still binds.
- **Sponsor goes silent:** 10-day acceptance window, then auto-accept or mediator.
- **AI wrote most of it:** credit to the human owner, AI share shown on the receipt.
- **Unfair rejection:** must cite a criterion; dispute with ledger evidence, escrow frozen.
- **Paid becomes unpaid:** new charter version; re-accept or leave with credit.

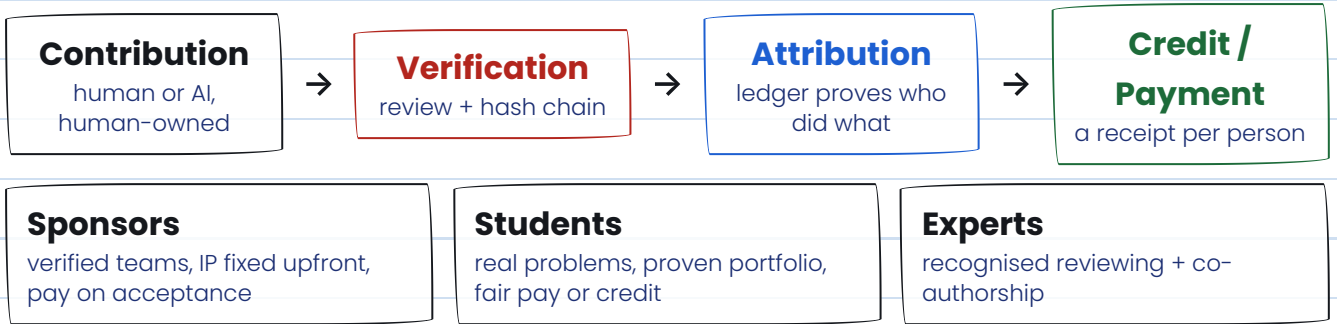
10 Focus area, build plan & tech stack

Who	Builds
[Name 1]	Screens, login, projects, charter
[Name 2]	Ledger, escrow, charter engine
[Name 3]	AI agents, matching, integrity check

3–10 PM spine · **10 PM–4 AM** engine + agents · **4–10 AM** corner cases · **10 AM–3 PM** test, README, video
Cut first: matching extras. **Never cut:** ledger + engine.

Focus	C · Fair rewards
Stack	Next.js · Supabase · Vercel · SHA-256
LLM	Gemini Flash (free) or Llama via Groq
Mocked	payments, KYC, similarity → real services later

11 Summary



12 What makes us different

- **Charter = code:** the terms people sign are the rules the engine runs.
- **Receipts, not totals:** every rupee traces back to a rule + reviewed work.
- **Corner cases are events:** log it, and the engine re-runs live.
- **AI with a leash:** agents draft, humans approve and get the credit.

13 Ready to scale

Today (prototype)	Tomorrow (real use)
Simulated escrow	Licensed payment partner
Synthetic KYC	Government-ID verification
Mocked similarity check	Real plagiarism + AI detector
One ledger chain	Hash anchored on a public blockchain

Costs grow with use: AI is paid from each project's 5% reserve.

Terms that run as code. Every rupee has a receipt.